Retail Orders Analysis

Python (pandas) and SQL (SQLite) analysis of 9,994 retail orders, 2022 to 2023

Gaurav Kalamkhede · October 2026

1. Summary

This project cleans a raw retail orders file with pandas, loads it into SQLite and answers business questions in SQL. It follows the flow of Ankit Bansal's tutorial *End to End Data Analytics Project (Python + SQL)*, then adds window-function analysis, an Excel dashboard and this report.

2,215,859 Total sales	**205,169** Total profit	**9.26%** Profit margin
9,994 Orders	**221.72** Average order value	**+2.25%** Sales growth 2023 vs 2022

Measure	Value
Period	2022-01-01 to 2023-12-31
Products	1,862
Total discount given	80,331
Sales 2022 / 2023	1,095,590 / 1,120,268
Profit 2022 / 2023	101,620 / 103,548
Profit growth 2023 vs 2022	+1.90%

Key findings

1. Sales grew 2.25% in 2023 (1,095,590 to 1,120,268) and profit grew 1.90%. The overall margin is 9.26%.
2. One copier, TEC-CO-10004722, is the top product with 59,514 in sales, more than twice the next product. It ranks first in three of the four regions.
3. Monthly sales are uneven: February 2023 was 42% above February 2022, but June 2023 was 27% below June 2022.
4. Machines had the highest profit growth (+3,635, +50%). Copiers, Appliances and Tables lost the most profit.
5. The South has the best margin (10.06%) but the lowest sales. The West sells the most, but its sales fell 11.2% in 2023.
6. Margin falls as discount rises: 10.54% at a 2% discount, 7.63% at 5%.
7. 40% of order lines lose money because the item has no markup and still gets a discount. The total loss is small (about 3.2k), but it is easy to stop.

2. Data and cleaning

Source: Kaggle *Retail Orders* (ankitbansal06/retail-orders), file `orders.csv`: 9,994 rows and 16 columns, all orders in the United States. Each row is one order (order_id is unique). `list_price` is the line total before discount.

Step	Why
Read with `na_values=['Not Available','unknown']`	Missing ship modes were written as words, so pandas would count them as real ship modes.
Rename columns to snake_case	Headers had spaces and mixed case ('cost price', 'Order Id'), which are awkward in pandas and SQL.
Derive discount, sale_price, profit	The file has list price, cost and discount % only. Revenue and profit must be calculated.
Convert order_date to a date	It was read as text, so year and month could not be extracted reliably.
Drop list_price and cost_price	They are captured by the new columns. discount_percent is kept for the discount analysis.

```
In [3]: df = pd.read_csv(RAW, na_values=["Not Available", "unknown"])
        print(df.shape)
        print(df["Ship Mode"].value_counts(dropna=False))
        df.head()
```

```
(9994, 16)
Ship Mode
Standard Class    5962
Second Class      1945
First Class       1538
Same Day           543
NaN                  6
Name: count, dtype: int64
```

Out[3]:

	Order Id	Order Date	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub Category	Product Id	cost price	List Price	Quantity	Disc Per
0	1	2023-03-01	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	FUR-BO-10001798	240	260	2	
1	2	2023-08-15	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	FUR-CH-10000454	600	730	3	
2	3	2023-01-10	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Labels	OFF-LA-10000240	10	10	2	
3	4	2022-06-18	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	FUR-TA-10000577	780	960	5	
4	5	2022-07-13	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies	Storage	OFF-ST-10000760	20	20	2	

Figure 1. Notebook: reading the file with na_values turns 'Not Available' and 'unknown' into NaN (6 missing ship modes).

```
In [8]: checks = pd.DataFrame({
            "check": ["rows", "columns", "duplicate order_id", "missing values (all columns)",
                      "missing ship_mode", "negative profit lines", "zero-value lines (sale_price = 0)"],
            "before (raw)": [len(raw_default), raw_default.shape[1], raw_default["Order Id"].duplicated().sum(),
                             raw_default.isna().sum().sum(), raw_default["Ship Mode"].isna().sum(), "n/a", "n/a"],
            "after (clean)": [len(df), df.shape[1], df["order_id"].duplicated().sum(),
                              df.isna().sum().sum(), df["ship_mode"].isna().sum(),
                              (df["profit"] < 0).sum(), (df["sale_price"] == 0).sum()],
        })
        checks
```

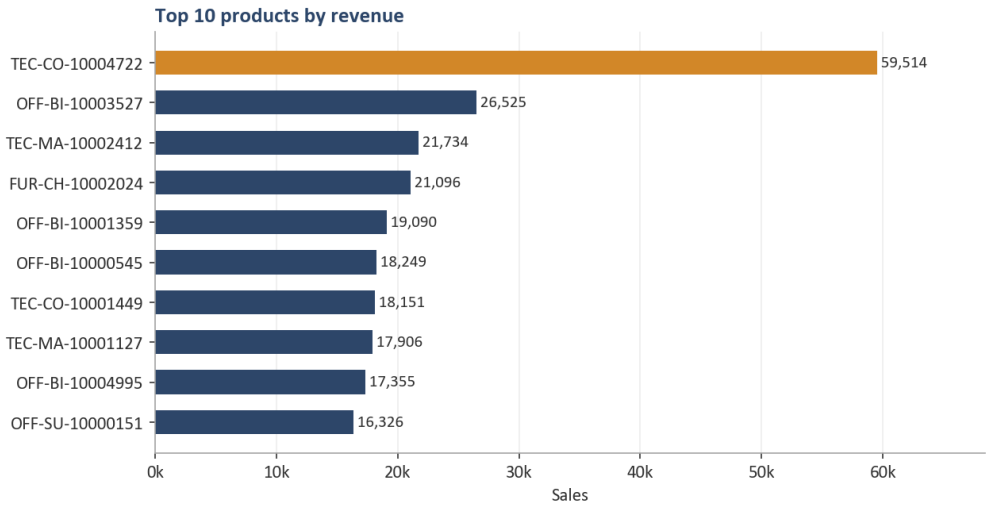
Out[8]:

	check	before (raw)	after (clean)
0	rows	9994	9994
1	columns	16	17
2	duplicate order_id	0	0
3	missing values (all columns)	1	6
4	missing ship_mode	1	6
5	negative profit lines	n/a	3997
6	zero-value lines (sale_price = 0)	n/a	507

Figure 2. Notebook: checks before and after cleaning. No rows are lost and there are no duplicate orders.

3. Business questions from the tutorial

Q1. Top 10 products by revenue



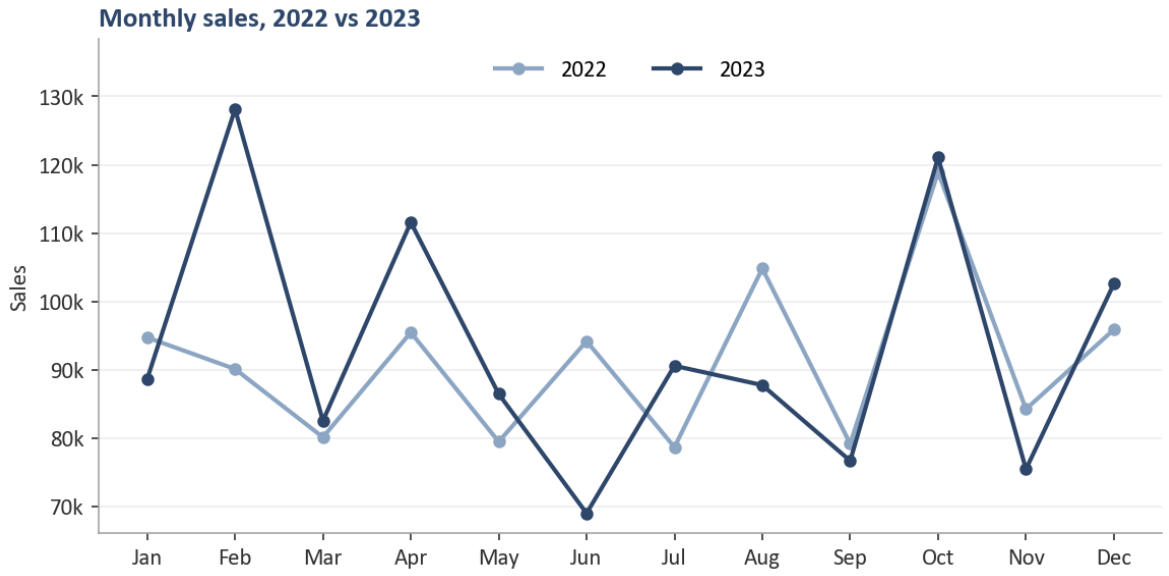
TEC-CO-10004722 earns 59,514, more than twice the second product. Four of the top ten are binders, and two are machines.

Q2. Top 5 products in each region

The same copier leads the Central, East and West regions. The South is led by a machine (TEC-MA-10002412).

Region	Rank	Product	Sales
Central	1	TEC-CO-10004722	16,975
Central	2	TEC-MA-10000822	13,770
Central	3	OFF-BI-10001120	11,056
Central	4	OFF-BI-10000545	10,133
Central	5	OFF-BI-10004995	8,416
East	1	TEC-CO-10004722	29,099
East	2	TEC-MA-10001047	13,767
East	3	FUR-BO-10004834	11,274
East	4	OFF-BI-10001359	8,464
East	5	TEC-CO-10001449	8,316
South	1	TEC-MA-10002412	21,734
South	2	TEC-MA-10001127	11,116
South	3	OFF-BI-10001359	8,053
South	4	TEC-MA-10004125	7,840
South	5	OFF-BI-10003527	7,391
West	1	TEC-CO-10004722	13,440
West	2	OFF-SU-10000151	12,592
West	3	FUR-CH-10001215	9,604
West	4	OFF-BI-10003527	7,805
West	5	TEC-AC-10003832	7,723

Q3. Month-over-month sales, 2022 vs 2023



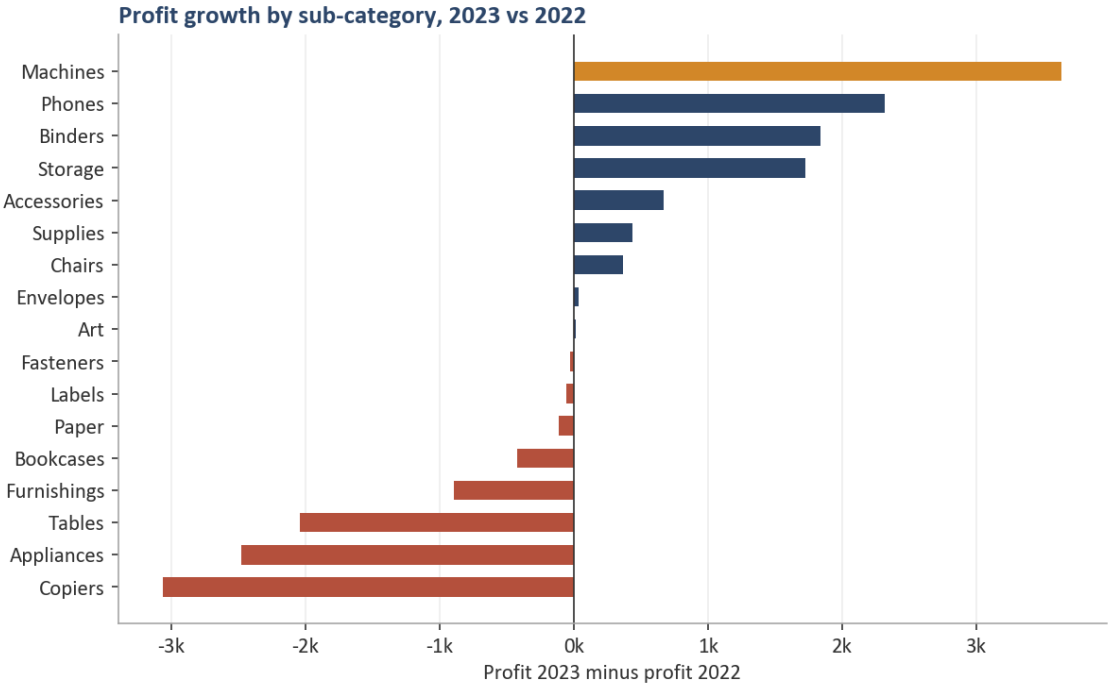
Month	Sales 2022	Sales 2023	Growth
1	94,712	88,633	-6.42%
2	90,091	128,124	42.22%
3	80,106	82,512	3.00%
4	95,452	111,569	16.88%
5	79,448	86,448	8.81%
6	94,170	68,976	-26.75%
7	78,652	90,564	15.14%
8	104,808	87,734	-16.29%
9	79,142	76,659	-3.14%
10	118,913	121,062	1.81%
11	84,225	75,433	-10.44%
12	95,870	102,556	6.97%

October was the best month of 2022 and February the best of 2023. 2023 beat 2022 in 7 of 12 months; February (+42%) and April (+17%) gained most, June (-27%) and August (-16%) fell most.

Q4. Highest-sales month for each category

Category	Best month	Sales
Furniture	2022-10	42,889
Office Supplies	2023-02	44,118
Technology	2023-10	53,000

Q5. Sub-category with the highest profit growth, 2023 vs 2022



Machines grew profit the most: from 7,243 to 10,878 (+50.19%). Nine sub-categories grew and eight declined. Copiers, Appliances and Tables together lost about 7.6k of profit.

```
In [17]: run("q5_subcategory_profit_growth")
```

```
-- Q5. Sub-categories by profit growth, 2023 vs 2022 (the first row answers the question).
```

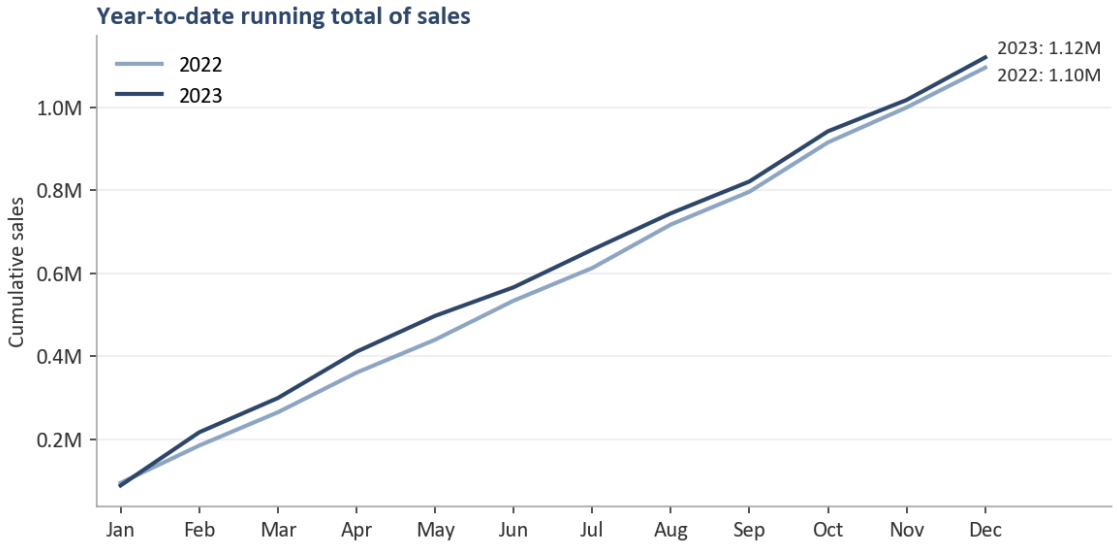
Out[17]:

	sub_category	profit_2022	profit_2023	profit_growth	growth_pct
0	Machines	7243.2	10878.5	3635.3	50.19
1	Phones	13024.7	15343.6	2318.9	17.80
2	Binders	8685.5	10523.1	1837.6	21.16
3	Storage	8907.4	10630.6	1723.2	19.35
4	Accessories	7387.2	8057.4	670.2	9.07
5	Supplies	1500.7	1937.4	436.7	29.10
6	Chairs	14725.3	15089.8	364.5	2.48
7	Envelopes	607.2	640.0	32.8	5.40
8	Art	924.1	935.7	11.6	1.26
9	Fasteners	40.1	8.8	-31.3	-78.05
10	Labels	349.6	295.3	-54.3	-15.53
11	Paper	3058.9	2942.4	-116.5	-3.81
12	Bookcases	5459.5	5036.6	-422.9	-7.75
13	Furnishings	4236.2	3342.1	-894.1	-21.11
14	Tables	10315.9	8275.3	-2040.6	-19.78
15	Appliances	6374.4	3893.3	-2481.1	-38.92
16	Copiers	8780.3	5718.6	-3061.7	-34.87

Figure 3. Notebook: the Q5 query result from SQLite.

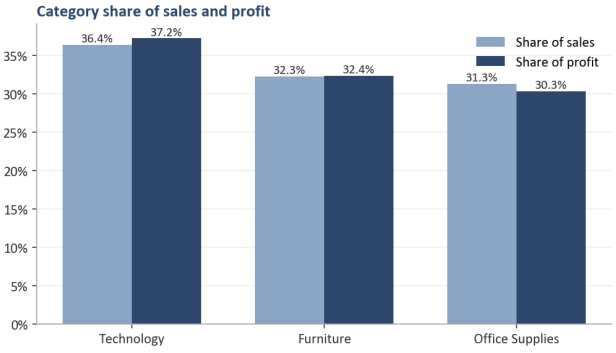
4. Extended analysis (window functions and CTEs)

Q6. Year-to-date running total



2023 was behind 2022 only in January, stayed ahead from February onwards, and finished 2.25% higher.

Q7. Category share of sales and profit



Category	Sales share	Profit share
Technology	36.41%	37.25%
Furniture	32.28%	32.40%
Office Supplies	31.30%	30.34%

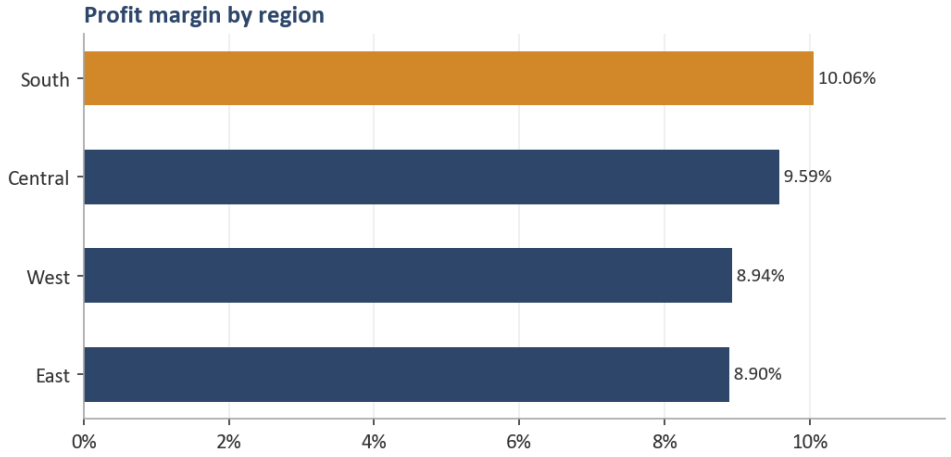
Technology is the largest category and earns a slightly larger share of profit than of sales. The three categories are close in size.

Q8. Sub-categories ranked within each category

Category	Sub-category	Sales	Profit	Margin	Rank
Furniture	Chairs	316,825	29,815	9.41%	1
Furniture	Tables	199,361	18,591	9.33%	2
Furniture	Bookcases	110,816	10,496	9.47%	3
Furniture	Furnishings	88,338	7,578	8.58%	4
Office Supplies	Storage	215,908	19,538	9.05%	1
Office Supplies	Binders	196,039	19,209	9.80%	2
Office Supplies	Appliances	103,698	10,268	9.90%	3
Office Supplies	Paper	75,831	6,001	7.91%	4
Office Supplies	Supplies	45,058	3,438	7.63%	5
Office Supplies	Art	26,260	1,860	7.08%	6
Office Supplies	Envelopes	15,917	1,247	7.84%	7
Office Supplies	Labels	11,995	645	5.38%	8
Office Supplies	Fasteners	2,939	49	1.66%	9
Technology	Phones	318,008	28,368	8.92%	1
Technology	Machines	182,902	18,122	9.91%	2
Technology	Accessories	161,605	15,445	9.56%	3
Technology	Copiers	144,359	14,499	10.04%	4

Chairs, Storage and Phones lead their categories. Fasteners (1.66%) and Labels (5.38%) have the thinnest margins.

Q9. Regional profitability



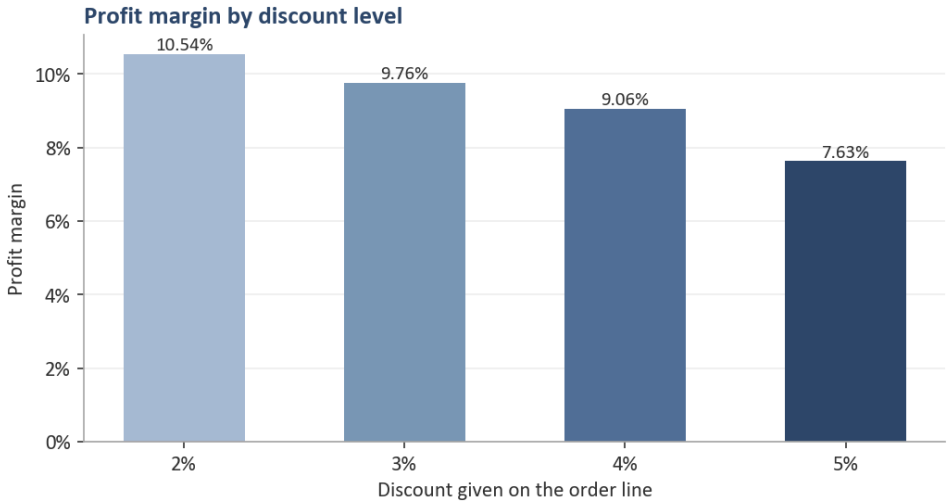
Region	Sales	Profit	Margin	Sales growth	Profit rank
West	699,859	62,559	8.94%	-11.20%	1
East	654,320	58,220	8.90%	11.06%	2
Central	483,669	46,379	9.59%	13.09%	3
South	378,012	38,012	10.06%	1.17%	4

Q10. Segment profitability

Segment	Orders	Sales	Profit	Margin	Avg order value
Consumer	5,191	1,119,696	101,586	9.07%	215.70
Corporate	3,020	681,658	65,638	9.63%	225.71
Home Office	1,783	414,505	37,945	9.15%	232.48

Consumers bring half of all sales. Corporate customers have the best margin (9.63%) and Home Office the largest average order.

Q11. Discount impact on profit



Discount %	Lines	Sales	Discount given	Profit	Margin
2	2,523	563,500	11,500	59,380	10.54%
3	2,542	563,483	17,427	54,993	9.76%
4	2,497	538,560	22,440	48,820	9.06%
5	2,432	550,316	28,964	41,976	7.63%

Each extra point of discount lowers the margin by roughly one point. The discount levels are spread evenly across lines, so this is not caused by a few large orders.

5. Recommendations

1. **Stop discounting items sold at cost.** Almost all loss-making lines have no markup. A simple rule (no discount when list price equals cost) removes the losses.
2. **Review the 4-5% discount levels.** They give about 51k of discount for a margin of 7.6-9.1%, against 10.5% at 2%.
3. **Look into the West.** It is the largest region, but sales fell 11.2% in 2023 while the other regions grew.
4. **Protect the top copier.** One product brings 2.7% of all sales. Keep it in stock and watch its pricing.
5. **Investigate falling sub-categories.** Copiers, Appliances and Tables lost the most profit. Check pricing, cost changes and product mix.
6. **Plan for the busy months.** October was one of the two best months in both years, and February and April were strong in 2023, so stock and staff can be planned ahead.

6. Method and tools

- Python 3.13 with pandas for cleaning, SQLite for the SQL analysis (all queries in `sql/analysis.sql`).
- Jupyter notebook for the walkthrough; `src/pipeline.py` reproduces every output in one run.
- matplotlib for charts, openpyxl for the Excel dashboard, Playwright with Microsoft Edge for this PDF, python-pptx for the slides.
- Totals were checked to match between SQL and pandas.

```
In [25]: sql_totals = pd.read_sql_query("SELECT ROUND(SUM(sale_price), 2) AS sales, ROUND(SUM(profit), 2) AS profit FROM df_orders", conn)
         assert sql_totals.loc[0, "sales"] == kpis["total_sales"]
         assert sql_totals.loc[0, "profit"] == kpis["total_profit"]
         print("SQL and pandas totals match:", sql_totals.to_dict("records")[0])

         SQL and pandas totals match: {'sales': 2215858.7, 'profit': 205168.7}
```

Figure 4. Notebook: SQL and pandas totals match.

Acknowledgement

The project flow and the five core business questions follow Ankit Bansal's YouTube tutorial *End to End Data Analytics Project (Python + SQL)* (youtube.com/watch?v=uL0-6kfiH3g) and his Kaggle dataset. The code was written fresh for this project, and the extended analysis, dashboard, report and slides were added.